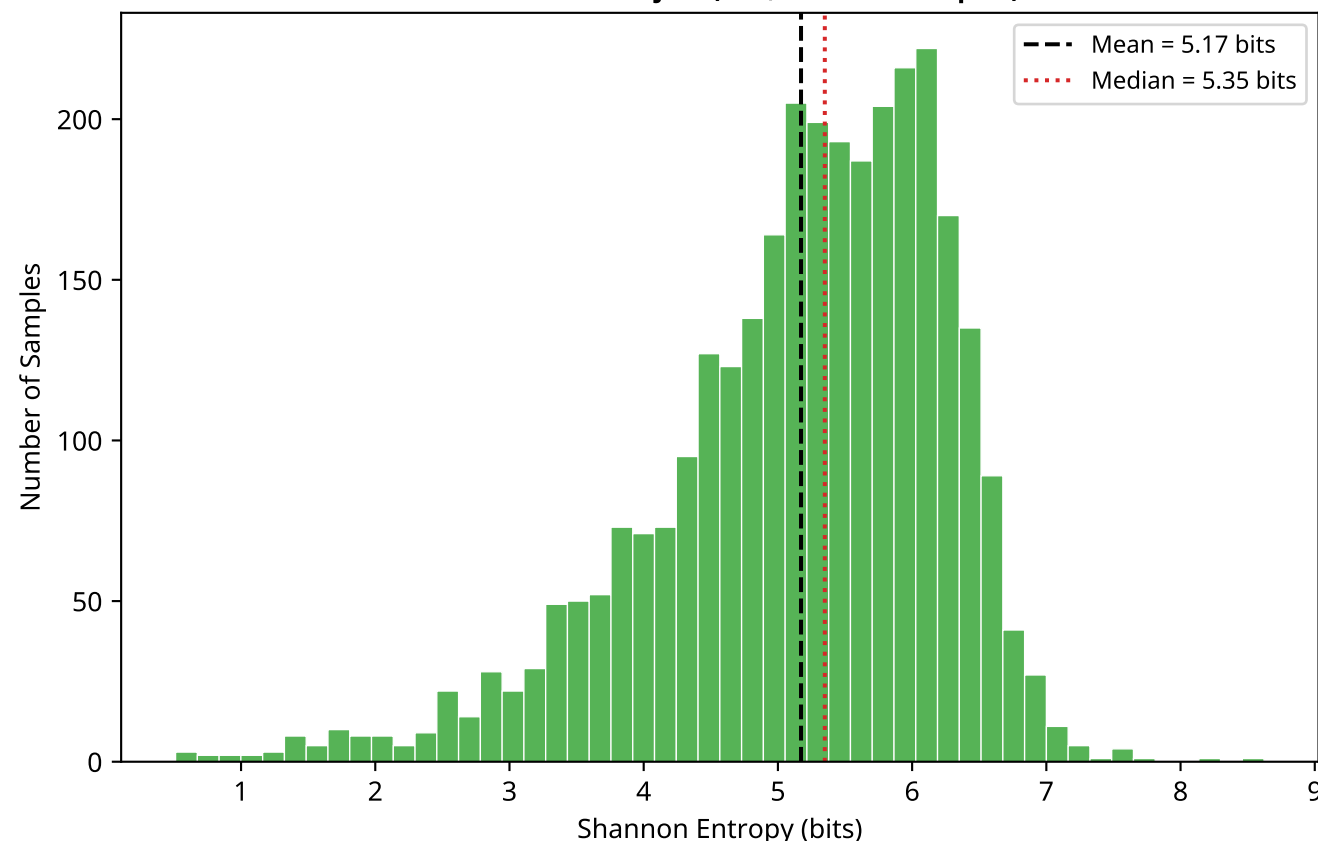
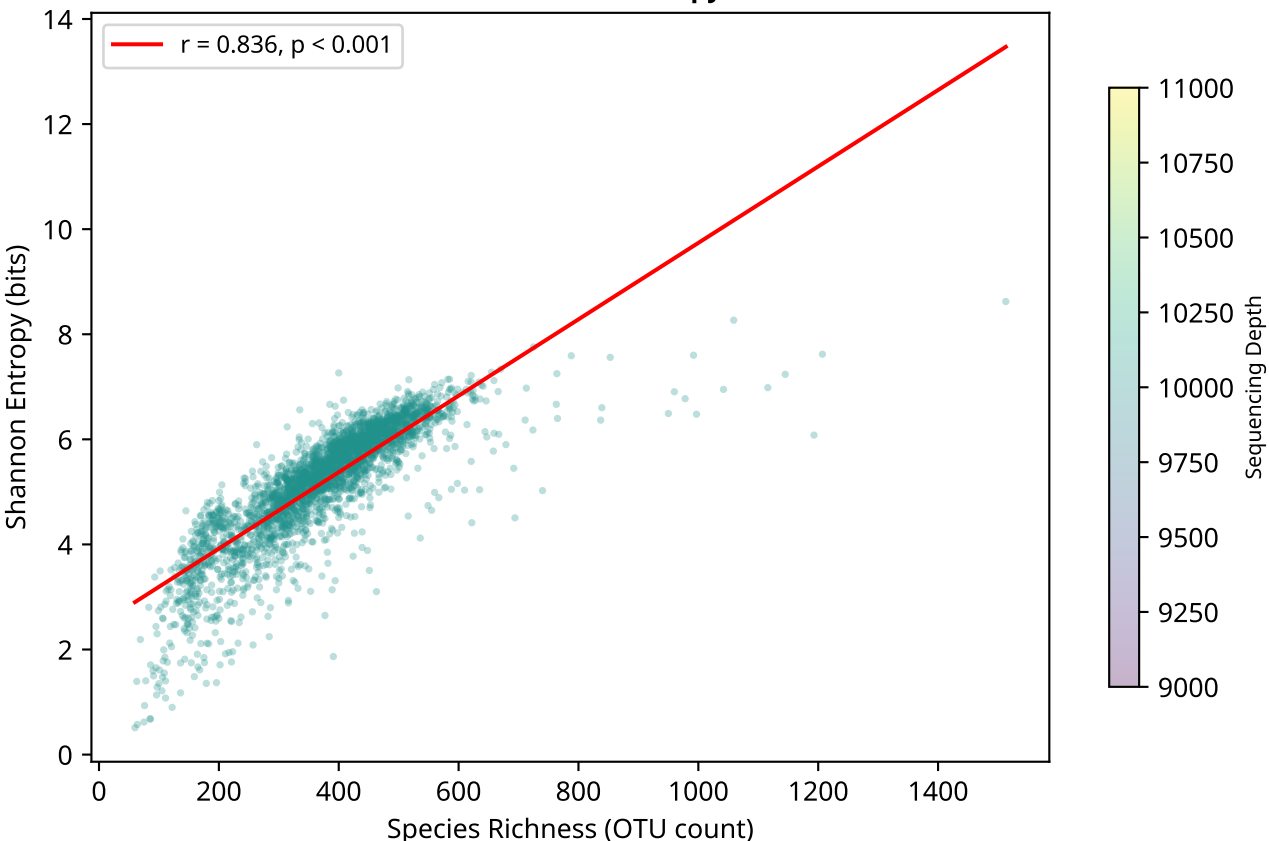


Figure 4: Entropy Analysis Across Biological Scales

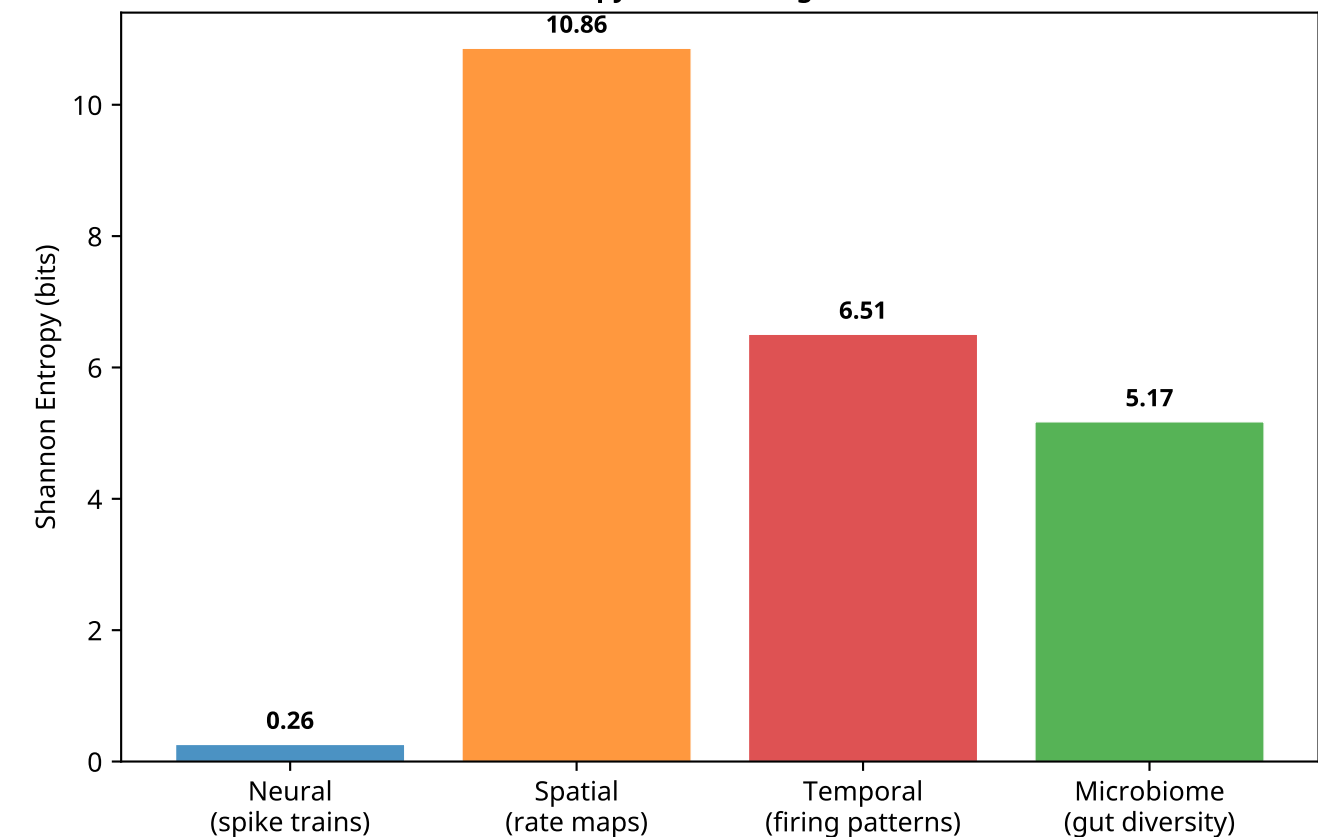
A. Gut Microbiome Shannon Entropy Distribution
American Gut Project (n=3,107 stool samples)



B. Richness vs. Entropy



C. Entropy Across Biological Scales



D. Multi-Scale Entropy Framework

